Systematic Literature Review of AI/ML Cost Estimation Methodology

Search String

This approach was intended to identify all paradigms of AI, focus on cost-oriented construction research, and encompass all significant types of construction projects. The search query was built using a conceptual block method: it was organized around three main conceptual blocks: (i) AI Block Identifies all paradigms (symbolic, ML, DL, hybrid) (ii) Cost Block Concentrates exclusively on cost paradigms (iii) Project Block Limits to construction/building context

The following Boolean search string was used for the search:

("artificial intelligence" OR AI OR "computational intelligence" OR "machine intelligence" OR "machine learning" OR ML OR "deep learning" OR DL OR "neural networks" OR "natural language processing" OR "expert system" OR "knowledge-based system" OR "decision support system" OR "computer vision" OR robotics OR "business intelligence")

AND ("cost estimation" OR "cost estimating" OR "construction cost" OR "project cost" OR "cost forecasting" OR "cost modelling" OR "construction estimating")

AND ("construction project" OR "building project*" OR "infrastructure project*" OR "residential project*" OR "commercial building" OR "institutional building" OR "industrial building")*

The phrase "machine learning" was further expanded to include symbolic, hybrid, and data-driven paradigms in the AI block. Although neural networks, deep learning, and ensemble methods are data-driven paradigms, expert systems and decision support systems are symbolic AI from the early days (Pan & Zhang, 2021; Baduge et al., 2022). The independence of paradigms is guaranteed by including all these phrases.
Only those phrases that have a direct relevance to forecasting and estimation were considered in the cost block. Following the previous cost modeling review studies by Kim et al. (2013) and Cheng et al. (2018), the general phrases such as "project performance" or "cost control" include non-estimation research, which reduces the accuracy of analysis. Hence, only phrases that explicitly state modeling or cost prediction tasks were considered.
The project block reflects the functional typologies commonly used in construction research (residential, commercial, institutional, industrial, and infrastructure), as recommended by

Baduge et al. (2022) and Darko et al. (2020). This ensures that AI techniques are analysed within context-specific project environments, rather than as generic computational tools.

Title and Abstract Screening

The first screening stage involved titles and abstracts review the of all exported papers. At this stage, three exclusion criteria were defined, the studies were excluded if they:

i. Did not use artificial intelligence or machine learning techniques,
ii. Did not focus on cost estimation, forecasting, or modelling,
iii. Were conducted in non-construction domains

Also three inclusion criteria were defined at this stage, for a paper to be included it is required that:

i. Artificial intelligence or machine learning are the primary analytical approach,
ii. Cost estimation was treated as a core dependent variable, and

The study was situated within a building or construction project context.

The screening and evaluation of the search results against resulted in reducing the dataset from 2,645 to 47 studies. The purpose of this stage was to remove irrelevant literature while retaining studies that were used for the review, consistent with the approach adopted in prior SLRs in construction informatics (Pan & Zhang, 2021; Baduge et al., 2022; Wuni et al 2019; Wuni 2024).

Full-Text Assessment

An extraction method was employed for the full-text analysis. In an attempt to minimize researcher bias, enhance replicability, and ensure consistency across experiments, a 17-field extraction procedure was developed. A simple bibliographic extraction is not sufficient for areas such as AI, which involve organizational, technical, and human factors.
In an attempt to ensure that the review encompasses the technical aspects of AI models, the project and cost estimation scenario, and the organizational, human, and environmental factors of adoption, the 17-field extraction method was developed by combining existing systematic review studies (Wuni et al 2019, Wuni 2024, Wuni 2025).

The Technical characteristics of AI models comprised of the following fields: AI Technique; Methodology; Evaluation/validation method; Research findings; Limitations. The Project and

cost estimation context comprised of the following: Building types; Cost Estimation Aspects; Country; Research Aim. While the Critical Success factors, Risks Factors and Skills are the fields included in the Organisational, human, and environmental adoption factors.

Field

Authors

Year

Title

Journal

DOI

Country

Research Aim

Methodology

Evaluation Method

Findings

Limitations

Building Type

AI Technique

Cost Estimation Aspect

CSFs

Risks

Skills

PRISMA Summary of the Review Process

The overall screening and selection procedure is summarised in accordance with the PRISMA framework. Although a flow diagram is provided separately, the numerical summary is as follows:

- Records identified through database searching: 2,645
- Studies included in the final review: 47

This PRISMA-based reporting ensures transparency, replicability, and methodological rigour, as recommended in contemporary systematic review standards. More importantly, it demonstrates that the final sample was not arbitrarily selected, but emerged through a structured and defensible filtering process.

Result of the Study

Building types Identified

With 55.3% of all studies categorized under general construction projects, Table 1 above shows a clear focus of AI cost estimating studies on generic projects. This indicates that most of the current AI cost estimating models have been developed using datasets that do not define functional project types or are a mix of several types of construction projects. While this is very useful for generalizability, it has also caused the model predictions to be less sensitive to the project context. The dominance of generalized datasets is a serious concern for the external validity of the models in specific project contexts, as the cost behavior of construction projects is actually very sensitive to project function, size, and complexity.

Table 1: Building types Identified

Building Type	No	Percentage (%)
General building projects	26	55.30%
Residential buildings	7	14.90%
High-rise / tall buildings	6	12.80%
Green / sustainable buildings	4	8.50%

Specialised building systems	4	8.50%
Total	47	100%

Residential buildings represent the second most frequently studied category, accounting for 14.9% of the reviewed literature. This relatively higher representation can be attributed to the repetitive nature of residential construction, standardised design templates, and the availability of large historical datasets. From a modelling perspective, residential projects present more stable cost drivers and reduced design variability, making them attractive for data-driven learning techniques. However, the modest share of residential-focused studies still indicates that sector-specific AI cost models remain limited, despite their potential for higher predictive reliability.

High-rise and tall buildings account for only 12.8% of the reviewed studies. This is particularly notable given that such projects are among the most complex and cost-volatile in the construction industry. Their under-representation suggests that researchers may be constrained by the scarcity of high-quality datasets, proprietary cost information, and the multidimensional risk profiles associated with vertical structures. Consequently, AI applications in this domain remain exploratory rather than mature, highlighting a critical gap in the literature where intelligent cost prediction tools could yield the greatest value.

Similarly, green and sustainable buildings constitute only 8.5% of the sample. This low proportion is paradoxical, considering the growing global emphasis on sustainability and the inherent cost uncertainty associated with green technologies, life-cycle considerations, and regulatory compliance. The limited focus on this category indicates that current AI cost estimation research has not yet aligned with emerging sustainability priorities. This misalignment reveals a strategic opportunity for future studies to integrate AI with sustainability-driven cost modelling, particularly for early-stage decision support.

Specialised building systems, including structural systems, foundations, and façades, also account for 8.5% of the reviewed studies. These system-level investigations demonstrate the potential of AI for micro-cost intelligence and component-level optimisation. However, their limited presence suggests that the field remains predominantly project-centric, with

insufficient attention to the granular cost drivers that collectively determine overall project budgets.

The findings indicate that AI-based cost estimation research is currently methodologically broad but contextually shallow. The dominance of general building categories reflects an emphasis on model generalisation at the expense of contextual precision. This pattern aligns with the early maturity stage of a technological domain, where exploratory modelling precedes domain specialisation.

Cost Estimation Aspect

Table 2: Cost estimation aspects

Cost Aspect Group	Frequency	Percentage (%)
General / Total Project Cost	19	40.40%
Early / Conceptual Cost	12	25.50%
Detailed / Final Cost	4	8.50%
Component / System-Level Cost	9	19.10%
Life-Cycle Cost	3	6.40%
Total	47	100%

The majority of research (40.4%) is concerned with general or total project cost, where the cost scope is not specifically related to a defined project cost stage. This shows that most AI models are designed and tested independently of cost maturity models, thus hindering their applicability. In professional cost management, estimates are stage-dependent. The lack of alignment here shows that most AI applications are still conceptually removed from practical decision-making processes.

The second most popular category is early/conceptual cost estimation (25.5%), which relates to AACE Classes 4–5. This is because there is a growing appreciation of the use of AI in front-end decision support, where the degree of uncertainty is high and parametric models are

less effective. However, the fact that only one-quarter of studies explicitly target this critical stage indicates that the potential of AI to support early investment decisions remains under-exploited.

In contrast, detailed/final cost estimation (8.5%) is severely under-represented. This is particularly notable because tender-stage and execution-stage estimates are data-rich and operationally critical. The scarcity of AI applications at this stage suggests the presence of institutional barriers such as data confidentiality, organisational resistance, and system integration challenges. Consequently, AI remains marginal in core commercial decision-making processes.

Component and system-level cost estimation accounts for 19.1% of the reviewed studies. These investigations focus on structural systems, foundations, green components, and material quantities. While this indicates emerging interest in micro-level cost intelligence, the limited scale of such studies suggests that AI research remains largely project-centric rather than process- or component-oriented.

Finally, life-cycle and risk-adjusted cost estimation represents only 6.4% of the sample. This reveals a significant disconnect between AI cost modelling research and the growing emphasis on sustainability, asset management, and whole-life value optimisation. Despite global calls for life-cycle thinking, AI applications remain heavily concentrated on short-term project costs.

AI-based cost estimation research is functionally misaligned with professional cost management practice. The dominance of generic cost targets and the marginal attention to life-cycle and detailed-stage costs indicate that current models prioritise algorithmic experimentation over decision relevance.

AI Techniques

Table 3:AI Techniques

AI Model Family	Frequency	Percentage (%)
Hybrid / Optimised	20	42.60%

Tree / Kernel Machine Learning	12	25.50%
Neural Network–Based	10	21.30%
Deep Learning	3	6.38%
Statistical / Regression	2	4.26%
Total	47	100%

The results show that hybrid and optimised models now dominate the field, accounting for 42.6% of all reviewed studies. These models integrate two or more techniques (e.g., ANN combined with optimisation algorithms, ensemble learning, or fuzzy systems) to overcome the weaknesses of single algorithms such as overfitting, local minima, and slow convergence.

The second largest group comprises tree-based and kernel machine learning models (25.5%), including Random Forest, Gradient Boosting, Support Vector Machines, and XGBoost. These models are particularly well-suited for structured construction cost data and offer greater transparency than neural networks. Their growing presence suggests an emerging recognition of the need for robust and interpretable models in cost estimation.

Standalone neural network–based models represent 21.3% of the sample. Although ANN remains a foundational technique, its reduced dominance compared with hybrid and ensemble systems indicates that researchers are increasingly seeking performance enhancement through integration, rather than relying on conventional neural architectures alone.

In contrast, deep learning models account for only 6.4% of studies. Despite their success in other data-intensive fields, their limited adoption in construction cost estimation reflects constraints related to data volume, computational cost, and the predominance of tabular cost data rather than image- or text-based inputs.

Lastly, statistical and regression-based models represent only 4.25% of the literature. This confirms a clear shift away from purely parametric approaches toward data-driven intelligence, although it also raises concerns about declining model interpretability and theoretical transparency.

AI-based cost estimation research has transitioned from single-algorithm modelling to integrated hybrid systems, while simultaneously expanding the use of tree-based and kernel learning methods.

Your evidence now supports a hybrid or ensemble strategy benchmarked against tree-based models, rather than relying solely on traditional ANN.

Relationship Between Building Types and AI Model Families in Construction Cost Estimation

One of the key objectives of this review was to determine whether the selection of artificial intelligence (AI) techniques for construction cost estimation is influenced by the functional type of building projects. Previous studies have argued that project characteristics such as scale, complexity, and structural form significantly affect cost behaviour and data patterns, which in turn should inform the choice of predictive models. However, empirical evidence linking building types to model families remains limited.

Table 4: AI Technique x Building Types matrix

Building Depth / Context	**Tree / Kernel ML**	**Neural Networks**	**Hybrid / Optimised**	**Deep Learning**	**Statistical / Regression**	**Row Total**
General building projects	7	8	9	1	1	26
Residential buildings	2	1	3	1	0	7
High-rise / tall buildings	1	2	3	0	0	6
Green / sustainable buildings	1	0	1	1	1	4
Specialised building systems	1	0	4	0	0	4
Total	12	10	20	3	2	47

For general building projects (n = 26), all model families are represented, with a dominance of hybrid/optimised models (9) and neural networks (8), followed by tree/kernel methods (7). This pattern suggests that generic building datasets are used as experimental platforms where researchers test a wide range of AI approaches. The relatively balanced distribution indicates no single methodological paradigm dominates in general contexts.

In residential buildings (n = 7), hybrid models (3) form the dominant group, while tree/kernel and neural models appear less frequently. This reflects the structural and market-driven complexity of housing projects, where single algorithms may be insufficient to capture price variability, encouraging the use of integrated and ensemble approaches.

For high-rise and tall buildings (n = 6), the concentration of hybrid (3) and neural (2) models indicates a preference for nonlinear learning systems capable of modelling height-dependent cost drivers and structural interactions. Tree-based methods are minimally applied, reinforcing the view that vertically complex projects require more expressive modelling techniques.

Green and sustainable buildings (n = 4) exhibit a balanced reliance on hybrid (1), tree/kernel (1), deep learning (1), and statistical (1) models. This diversity reflects both data scarcity and methodological experimentation in sustainability-oriented cost modelling, where researchers are still exploring suitable analytical approaches.

Finally, specialised building systems (n = 4) show a strong dominance of hybrid/optimised models (4). This highlights the micro-level complexity of components such as foundations, structural systems, and façades, where multiple data sources and nonlinear cost relationships necessitate integrated modelling strategies.

AI model selection in construction cost estimation is strongly context-driven. As project complexity increases from general buildings to residential, high-rise, green, and component-level systems, the literature demonstrates a progressive shift toward hybrid and optimised AI models.

This indicates that future AI cost estimation systems, particularly for housing and complex building projects, should prioritise hybrid or ensemble architectures, benchmarked against neural and tree-based models, rather than relying on single-algorithm approaches.

This conclusion directly supports your research direction in developing an AI cost estimation model for housing projects.

Machine Learning Development

1. Conceptual Foundations of Machine Learning

Machine learning (ML) is a scientific discipline within artificial intelligence that focuses on developing algorithms capable of learning patterns from data and improving performance over time without explicit rule-based programming. Rather than encoding deterministic instructions, ML systems infer functional relationships directly from empirical observations (Alpaydin, 2014).

A widely accepted formal definition stated that:

A computer program learns from experience E with respect to some task T and performance measure P if its performance improves with experience (Mitchell, Sadiku et al., 2017).

This definition emphasizes three core elements:

- **Experience (data)**
- **Task (prediction or decision)**
- **Performance measure (accuracy or error)**

Machine learning therefore transforms raw data into predictive models through statistical optimization and computational learning processes.

Raw Data → Learning Algorithm → Model → Prediction/Decision

2. Machine Learning vs Traditional Rule-Based Programming

Traditional programming relies on explicit rules defined by programmers. The logic flow is deterministic:

Rules + Input Data → Output

Machine learning reverses this paradigm:

Input Data + Desired Output → Learning Algorithm → Learned Rules

Instead of manually specifying rules, ML extracts them automatically from examples (Sadiku et al., 2017).

This distinction is critical because many real-world systems exhibit nonlinear and high-dimensional relationships that are difficult to encode manually. ML provides adaptive mechanisms capable of generalization beyond observed data (Alpaydin, 2014).

3. Learning Paradigms in Machine Learning

Machine learning is organized into four primary paradigms based on how supervision is provided during learning.

3.1 Supervised Learning

Supervised learning uses labeled datasets where each input corresponds to a known output. The objective is to approximate a function:

$f(x) \rightarrow y$

Two major supervised tasks exist:

- **Regression**: predicting continuous values
- **Classification**: predicting discrete categories

The model minimizes prediction error by adjusting internal parameters during training (Praba, 2021). Supervised learning is the most widely applied paradigm in predictive modeling.

3.2 Unsupervised Learning

Unsupervised learning operates without labeled outputs. Instead, the algorithm discovers hidden structures or latent patterns in data.

Common objectives include:

- clustering
- dimensionality reduction
- association discovery

The goal is exploratory pattern recognition rather than direct prediction (Nguyen, 2025).

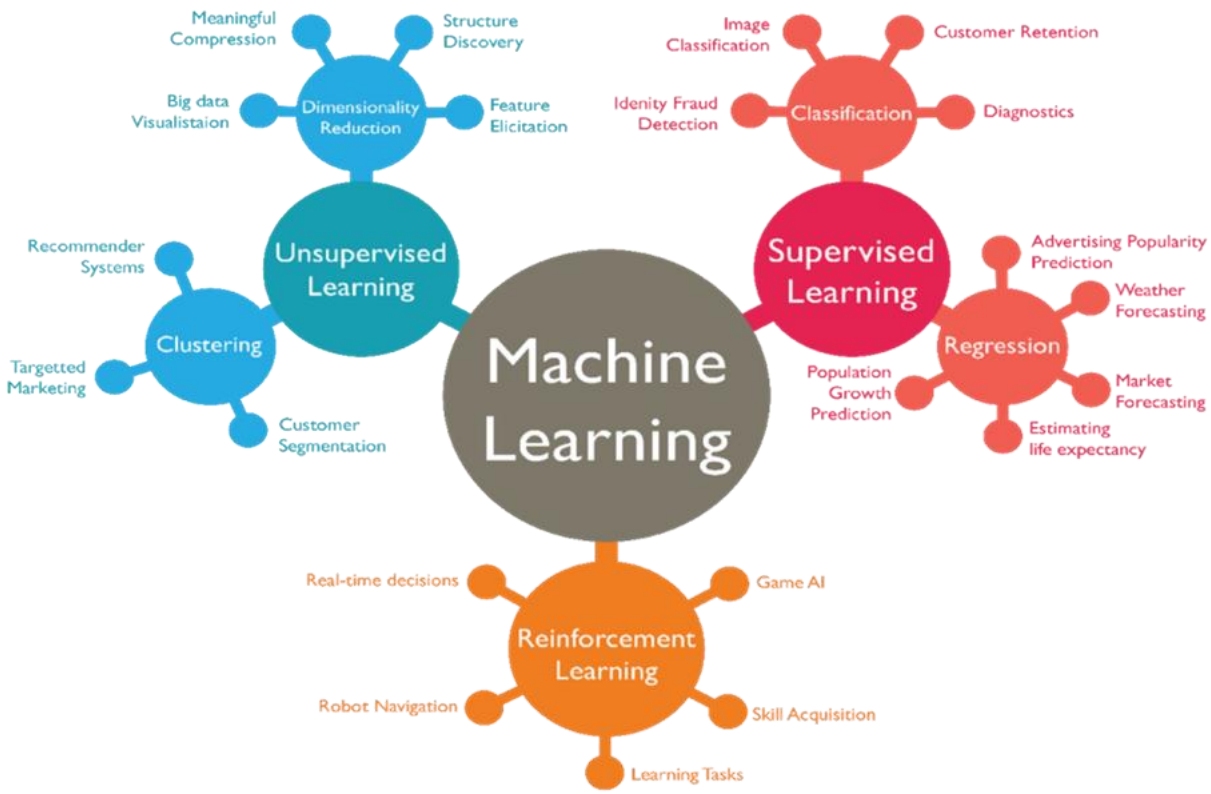
3.3 Semi-Supervised Learning

Semi-supervised learning combines limited labeled data with large volumes of unlabeled data. This approach is useful when labeling is expensive or impractical (Praba, 2021). It bridges the gap between supervised and unsupervised methods.

3.4 Reinforcement Learning

Reinforcement learning focuses on sequential decision-making. An agent interacts with an environment and receives feedback through rewards or penalties.

The agent learns optimal strategies by maximizing cumulative reward over time (Alpaydin, 2014).



4. Major Model Families in Machine Learning

Machine learning algorithms are categorized into families based on their mathematical structure.

4.1 Statistical Models

These models assume linear or parametric relationships between variables. Examples include linear regression and logistic regression. They are interpretable and computationally efficient but may struggle with complex nonlinear data (Sadiku et al., 2017).

4.2 Tree-Based Models

Tree-based algorithms partition data into hierarchical decision structures. These models are flexible and capable of capturing nonlinear relationships while maintaining interpretability (Praba, 2021). Examples include; Decision trees; Random forests; Gradient boosting.

4.3 Neural Networks

Neural networks simulate interconnected processing units inspired by biological neurons. They consist of:

- input layers
- hidden layers
- output layers

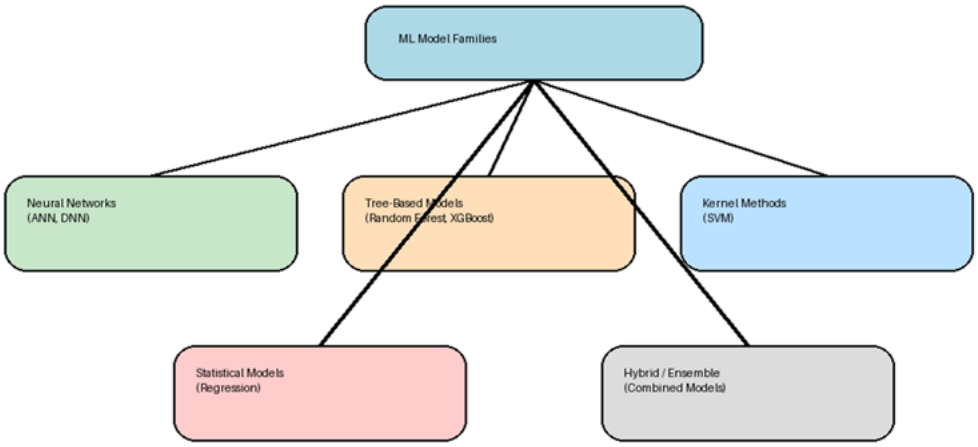
Through iterative optimization, neural networks approximate highly complex functions (Alpaydin, 2014).

4.4 Kernel Methods

Kernel-based algorithms, such as Support Vector Machines, map data into higher-dimensional spaces to improve separability. They are powerful for high-dimensional classification tasks (Nguyen, 2025).

4.5 Ensemble and Hybrid Models

Ensemble models combine multiple algorithms to enhance predictive performance. Hybrid models integrate different learning paradigms into unified systems. These approaches improve robustness and accuracy (Praba, 2021).



5. Data Representation in Machine Learning

Machine learning systems rely heavily on data representation.

Data types include:

- numerical variables
- categorical variables
- structured datasets
- Text data
- unstructured multimedia data
- Time-series data

Before training, data must be transformed into feature vectors suitable for algorithmic processing (Alpaydin, 2014).

<u>Development Process of Machine Learning Models</u>

The development of a machine learning model is a structured, iterative process that transforms raw data into a predictive or decision-making system. Although specific implementations vary

depending on the learning paradigm, the general workflow remains conceptually consistent across machine learning systems (Alpaydin, 2014; Sadiku et al., 2017).

The generic ML lifecycle consists of:

1. Problem definition
2. Data collection
3. Data preprocessing
4. Feature representation
5. Model selection
6. Model training
7. Model evaluation
8. Model optimization
9. Deployment and monitoring

This structured pipeline ensures that the model not only learns patterns from data but also generalizes effectively to new, unseen observations (Alpaydin, 2014).

2. General Machine Learning Development Framework

2.1 Problem Definition

The first stage involves clearly defining the learning task. According to Mitchell's formal definition, a learning system must specify:

- Task (T)
- Experience (E)
- Performance measure (P)

A clear formulation of these elements determines the type of learning paradigm to adopt (Sadiku et al., 2017).

For example:

- Regression task → predict continuous variable
- Classification task → assign categorical label
- Clustering task → identify latent groupings

2.2 Data Collection

Data serves as the foundation of learning. The quality, volume, and structure of the dataset directly influence model performance. Machine learning systems require representative and sufficiently large datasets to avoid bias and instability (Nguyen, 2025).

2.3 Data Preprocessing

Raw data often contains noise, missing values, inconsistencies, or irrelevant variables. Preprocessing may include:

- Cleaning
- Normalization or scaling
- Encoding categorical variables
- Removing outliers
- Handling imbalance

2.4 Feature Engineering and Representation

Feature representation plays a central role in ML performance (Alpaydin, 2014). Feature engineering transforms raw inputs into structured feature vectors suitable for algorithmic learning. Proper representation improves learning efficiency and predictive accuracy (Praba, 2021).

2.5 Model Selection

At this stage, a suitable model family is chosen based on:

- Nature of the task
- Data characteristics
- Computational requirements
- Interpretability needs

Model families include statistical models, tree-based models, neural networks, and kernel methods (Alpaydin, 2014).

2.6 Model Training

Training involves optimizing model parameters to minimize a loss function. This is achieved using mathematical optimization techniques such as gradient descent (Alpaydin, 2014).

2.7 Model Evaluation

Evaluation assesses generalization performance using separate validation data.

Common evaluation metrics include:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

- Accuracy
- Precision/Recall

Overfitting and underfitting are assessed at this stage (Sadiku et al., 2017).

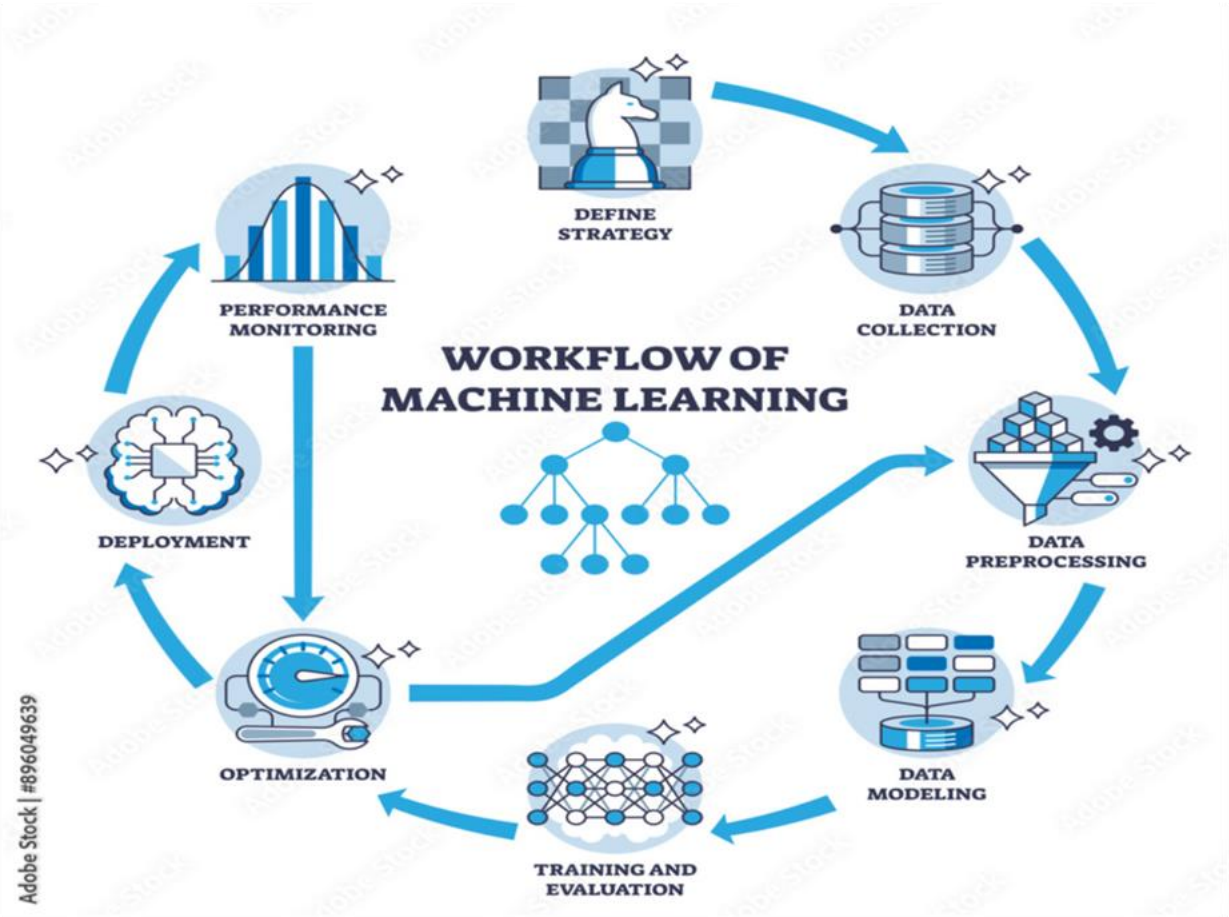
2.8 Model Optimization

Hyperparameter tuning improves model performance. Techniques include:

- Grid search
- Cross-validation
- Regularization (Praba, 2021)

2.9 Deployment and Monitoring

Once validated, the model is deployed into operational environments. Continuous monitoring ensures stability and adaptation to new data (Nguyen, 2025).



Research Methodology on Machine Learning Process

1. Problem Definition

The objective of this research is to develop a machine learning model capable of estimating the construction cost of housing projects in Nigeria during the early planning stage. Previous studies such as **Gunaydın & Doğan (2004)** and **Kim et al. (2005)** demonstrated that machine learning models can effectively predict construction costs by learning relationships between project characteristics and cost outcomes. Therefore, the present study defines the prediction problem as estimating **housing construction cost based on historical housing project data and relevant economic variables**.

2. Variable Selection

Feature representation involves selecting variables that influence construction cost. Previous studies have used a wide range of variables depending on the type of building project and the stage of cost estimation. **Gunaydın and Doğan (2004)** used structural design parameters such as building geometry and structural characteristics because these variables significantly affect structural system cost. Similarly, **Arafa and Alqedra (2011)** used variables such as building area, number of floors, number of apartments, foundation type, and finishing type because these parameters describe the physical characteristics of buildings that influence construction cost.

Other studies included additional project-level and economic variables. For instance, **Wang et al. (2024)** included variables such as labour cost level, material price level, construction duration, and project scale in their cost estimation model because these variables represent the economic and project conditions affecting construction cost. Similarly, **Al-Tawal et al. (2020)** used multiple design-stage variables to represent building systems, construction parameters, and site conditions.

Based on the literature, this research will represent housing construction cost using three groups of variables: **building construction parameters, project characteristics, and macroeconomic indicators**. Building parameters describe the physical characteristics of housing units, project characteristics represent project execution conditions, while macroeconomic indicators such as **labour cost index and material cost index** represent market conditions affecting construction costs.

Building Parameters

- Gross floor area
- Number of floors
- Number of housing units
- Structural system
- Foundation type

- Finishing type
- Building height

Project Characteristics

- Construction duration
- Project location
- Project scale

Macroeconomic Indicators

- Labour cost index
- Material cost index
- Inflation rate

3. Data Collection

Previous construction cost estimation studies have relied on **historical project data** because machine learning models require datasets containing both **input variables (project characteristics)** and **target variables (actual project cost)**. **Gunaydın and Doğan (2004)** developed their neural network model using data from completed residential building projects. Similarly, **Kim et al. (2005)** used a dataset of apartment housing projects to compare several predictive models, while **Arafa and Alqedra (2011)** used data from 71 building projects obtained from contractors and consulting offices. These studies demonstrate that historical project datasets allow machine learning algorithms to learn patterns linking building attributes and construction costs.

Based on these studies, this research will collect **historical housing project data from Nigeria**, including building characteristics, project parameters, and economic indicators together with the actual construction cost of each project. Using historical housing project data ensures that the machine learning models can learn realistic relationships between housing project characteristics and construction costs.

4. Data Pre-Processing

Construction project datasets usually contain variables with different units and scales, which can negatively affect the learning process of machine learning algorithms. Therefore, many previous studies perform **data preprocessing procedures** before model training. For example, **Arafa and Alqedra (2011)** applied **data normalization** before training their neural network model in order to transform all variables into comparable numerical ranges. This helps neural networks learn more efficiently because variables with large numerical values will not dominate the training process.

Similarly, **Chen et al. (2025)** performed **data cleaning and feature normalization** to remove inconsistent data entries and prepare the dataset for machine learning analysis. Data cleaning ensures that the dataset contains reliable information and reduces the influence of errors on model performance.

Other researchers have also applied **correlation analysis** before model training. For example, **Wang et al. (2024)** conducted correlation analysis between variables to understand which variables are strongly related to construction cost. This helps identify important cost drivers and remove redundant variables that may not contribute significantly to the prediction model.

In addition to data preparation, many studies divide datasets into subsets for model training and evaluation. For example, **Arafa and Alqedra (2011)** divided their dataset into **training and testing sets** so that the model could learn patterns from the training data and then be evaluated using unseen testing data. Similarly, **Bayram and Al-Jibouri (2016)** divided their dataset into **training and validation datasets** to ensure that the model was evaluated using independent data.

Based on these studies, the dataset in this research will undergo preprocessing procedures including **data cleaning, normalization, correlation analysis, and dataset partitioning into training and testing datasets**. These preprocessing steps are necessary to improve the reliability and predictive performance of the machine learning models.

5. Feature Selection (Data-Driven Selection)

After the dataset was collected and preprocessed, feature selection was performed to identify the most relevant predictors for the machine learning models. Feature selection is necessary because not all collected variables contribute equally to the prediction model, and the presence of redundant or irrelevant variables may reduce model accuracy and increase computational complexity.

Correlation analysis was first conducted to examine the relationship between each independent variable and the target variable (construction cost). Correlation analysis helps identify variables that have strong relationships with the dependent variable and detect highly correlated variables that may provide redundant information. Similar approaches have been applied in construction cost estimation studies to identify important cost drivers prior to model development (Wang et al., 2024).

In addition to correlation analysis, feature importance analysis was conducted using the Random Forest algorithm to evaluate the contribution of each variable to the prediction model. Random Forest is capable of estimating the relative importance of input variables based on their contribution to reducing prediction error during model training. Previous studies have demonstrated the usefulness of tree-based algorithms for identifying important predictors in construction cost estimation problems (Chen et al., 2025).

Variables that exhibited low correlation with the target variable or low importance scores in the Random Forest analysis were excluded from the final model. The remaining variables formed the final feature set used for training the machine learning models. This process ensures that the predictive models are trained using the most relevant and informative variables, thereby improving model accuracy and interpretability.

5. Model Selection

Different machine learning algorithms have been applied in construction cost estimation studies. Among these algorithms, **Artificial Neural Networks (ANN)** have been widely used because they can capture complex nonlinear relationships between project variables and construction cost. Previous Researches (**Gunaydın and Doğan 2004; Arafa and Alqedra 2011)** applied ANN to estimate residential building costs because neural networks can model nonlinear relationships between structural design parameters and construction cost. Similarly, used ANN to estimate building construction costs because neural networks can learn complex patterns from project datasets.

However, some studies have compared multiple algorithms to identify the most accurate predictive model. For instance, **Kim et al. (2005)** compared regression models, artificial neural networks, and case-based reasoning models to determine which technique produced the most accurate cost predictions. Similarly, **Chen et al. (2025)** compared several machine learning algorithms including ridge regression, K-nearest neighbour, and gradient boosting models to evaluate their predictive performance.

These studies indicate that comparing multiple algorithms allows researchers to identify the most suitable model for a given dataset. Therefore, following these studies, this research will develop and compare **Artificial Neural Network models and tree-based machine learning algorithms** in order to identify the model that provides the most accurate housing cost predictions. For the ANN, **Multi-Layer Perceptron (MLP) trained using the Back-Propagation learning algorithm** because the algorithm is capable of modelling complex nonlinear relationships between project parameters and construction cost. Random Forest will be selected for tree-based learning because it builds **multiple decision trees and aggregates their predictions**, which reduces the risk of overfitting and improves prediction stability. Previous studies such as **Chen et al. (2025)** and other construction cost estimation research have shown that ensemble tree-based algorithms often produce higher prediction accuracy compared with single decision tree models.

6. Model Training

Model training involves allowing the machine learning algorithm to learn relationships between project variables and construction cost outcomes. Previous studies have trained predictive models using historical construction project datasets. For example, **Gunaydın and Doğan (2004)** trained neural network models using structural design parameters and cost

data, allowing the model to learn relationships between structural characteristics and construction cost.

Similarly, **Arafa and Alqedra (2011)** trained neural networks using building project datasets collected from contractors and consultants. Training the model using historical project data allows the algorithm to identify patterns linking project characteristics and construction costs.

Following these studies, the models developed in this research will be trained using the **training portion of the dataset**, where housing project variables will be used as input variables and construction cost will be used as the target variable.

7. Model Evaluation

After model training, it is necessary to evaluate the performance of the predictive model. Previous studies have used several statistical metrics to measure prediction accuracy. For example, **Chen et al. (2025)** used **Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2)** to evaluate machine learning models for construction cost prediction. Similarly, **Bayram and Al-Jibouri (2016)** used **RMSE, MAE, and Mean Absolute Percentage Error (MAPE)** to evaluate the accuracy of cost prediction models.

These metrics measure the difference between predicted and actual project costs, allowing researchers to determine the reliability of the predictive models. Based on these studies, the predictive models developed in this research will be evaluated using **RMSE, MAPE, and R^2** to assess prediction accuracy.

8. Model Optimization

Model optimization is often used to improve prediction performance by adjusting model parameters or integrating optimization techniques. For example, **Wang et al. (2024)** improved the performance of neural network models by integrating optimization algorithms such as genetic algorithms and particle swarm optimization. These optimization techniques help identify the most suitable model parameters for improving prediction accuracy.

Similarly, other studies have applied feature analysis techniques such as **SHAP analysis** to improve model interpretability and understand the influence of different variables on construction cost predictions.

Based on these studies, model optimization in this research will involve adjusting algorithm parameters and evaluating different model configurations in order to identify the model that produces the most accurate housing cost predictions.

STAGE 1: Problem Definition

Cost estimation model

STAGE 2: Variables Identification

Building Parameters ↔ Project Characteristics

Project Characteristics ↕ Macroeconomic indicators

STAGE 3: Data Collection

Historical Project Data

STAGE 4: Data PreProcessing

Data Cleaning → Correlation Analysis

Correlation Analysis ↓ Data Normalization

Data Normalization → Train/Test Split

STAGE 5: Model Selection

MLP Neural Network ↔ Random Forest

STAGE 6: Model Evaluation

MAE → RMSE

$$\mathrm{RMSE} = \sqrt{\sum \frac{(y_{pred} - y_{ref})^2}{N}}$$

RMSE ↓ MAPE

MAPE → R2